

Foundations & Basics

Introduction to Databases: Concepts and Types



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What is database?

- Stores data in an organized way
- Easy to search, update, report and manage



Traditional Files



Database

Database vs. Traditional File System

Traditional File System Issues

- Duplication
- Inconsistency
- Slow search times

File System



Database

Name	Addr	Purchase	Email	Phone
Karzan Faiq	123 Abc St.	\$80	karzan@test.com	123.5269
Ari Ahmed	99 Hill Dr.	\$90	ari@test.com	125.5288
...	...	...	...	...
Rawaz Zrar	123 Oak St.	\$150	rawaz.z@example.com	555.1234

Types of Databases

- Relational
- NoSQL
- Cloud Databases



SQL



NoSQL



Cloud DB

Relational Databases (SQL)

- Organized into
 - Tables = entities
 - Columns = attributes
 - Rows = records/tuples
- Relationships connect data between tables
- Follow ACID rules for data integrity

OrderID	CustomerID	Order Date	Total_Amount
101	2	2025-09-10	150.00
102	1000	2025-09-12	678.95

CustomerID	Customer_Name	Email	Phone
1	Shakar Kochar	Shakar.k@test.com	123.5269
2	Sara Anwer	Sara@xyz.com	123.5269
...	...	...	...
1000	Alan Ali	Alan@example.com	555.1234

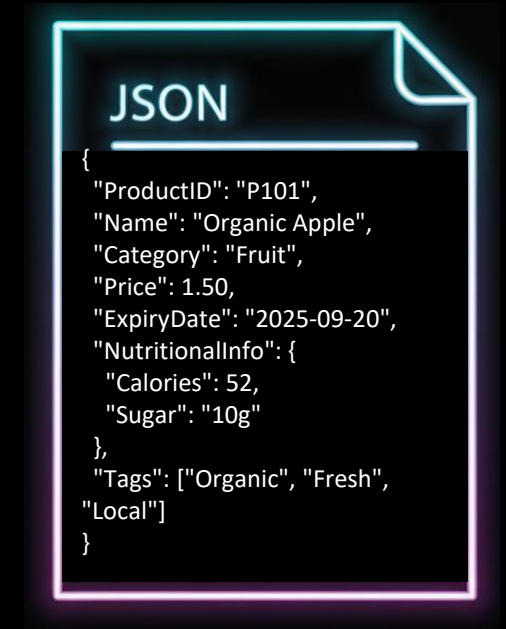
Row

Column

NoSQL and Cloud Databases

NoSQL

- Schema-less → No fixed structure required
- Great for flexible, evolving data
- Optimized for speed and scalability
- Common types: Document, Key-Value, Graph, Wide-Column



Cloud Databases

- Hosted online, accessible anywhere
- Managed services handle backups and scaling
- Pay only for what you use
- Great for dashboards and real-time reports



Azure, AWS, Google Cloud



Data



Cloud Accessible Anywhere

Coming Next

Database Design & Modeling

